Unit 12: Application layer – services and protocols

CONTENTS

Objectives

- understand Telnet, FTP
- understand E-mail, POP, IMAP
- understand domain name system
- understand WWW and HTTP

Introduction

Terminal NETwork is the abbreviation for Terminal Network. It's a protocol that allows one device to communicate with another on the same network. It is a basic TCP/IP protocol for virtual terminal services that is provided by ISO. The local machine is the one that initiates the communication. The remote computer is the computer to which the connection is made, i.e. the computer that recognises the connection. When the link between the local and remote computers is created. During a telnet session, whatever is going on on the remote computer is viewed on the local computer. Telnet works on a client/server model. The telnet client software is used on the local computer, while the telnet server programme is used on the remote computer.

12.1 Telnet Commands

A prefix character, Interpret As Command (IAC), with the code 255, is used to identify telnet commands. After IAC, there are command and choice codes. The command's basic format is seen in the Table 1 below.

Table 1Telnet Command Format

IAC	Command Code	Option Code

The Table 2 are some of the most commonly used TELNET commands:

Table 2 Telnet Commands

Character	Decimal	Binary	Meaning
WILL	251	11111011	1. Offering to enable. 2. Accepting a request to enable.
WON'T	252	11111100	1. Rejecting a request to enable. 2. Offering to disable. 3. Accepting a request to disable.
DO	253	11111101	1. Approving a request to enable. 2. Requesting to enable.
DON'T	254	11111110	1. Disapproving a request to enable. 2. Approving an offer to disable. 3. Requesting to disable.

The Table 3 are some commonly used telnet options:

Table 3 Telnet Options

Code	Option	Meaning
0	Binary	This translates to an 8-bit binary transmission.
1	Echo	It will repeat the information obtained from one hand to the other.
3	Suppress go ahead	After info, it will switch off the go-ahead signal.
5	Status	It will inquire about TELNET's position.
6	Timing mark	It establishes the timing markers.
8	Line Width	It determines the width of the rows.
9	Page Size	It determines how many lines are on a list.
24	Terminal type	It was used to specify the terminal type.

32	Terminal Speed	It regulated the terminal's pace.
34	Line Mode	It will switch to line mode.

Operational Modes:

Most telnet implementation operates in one of the following three modes :

- Default mode
- Character mode
- Line mode

Default Mode

- This mode is used if none of the other modes are invoked.
- In this mode, the client performs the echoing.
- In this mode, the user types a character, and the client enchoes it on the screen, but it does not submit it until the whole line is over.

Character Mode

- In this mode, each character typed by the client is sent to the server.
- In this mode, the server usually enchoes the character to be viewed on the client's computer.

Line Mode

- Line editing like echoing, character erasing etc is done from the client side.
- Client will send the whole line to the server.

12.2 File Transfer Mode(FTP)

FTP (File Transfer Protocol) is an application layer protocol that allows you to transfer files between local and remote file systems. It, like HTTP, runs on top of TCP. FTP uses two TCP links in parallel to pass a file: a control link and a data connection as seen in Figure 1.

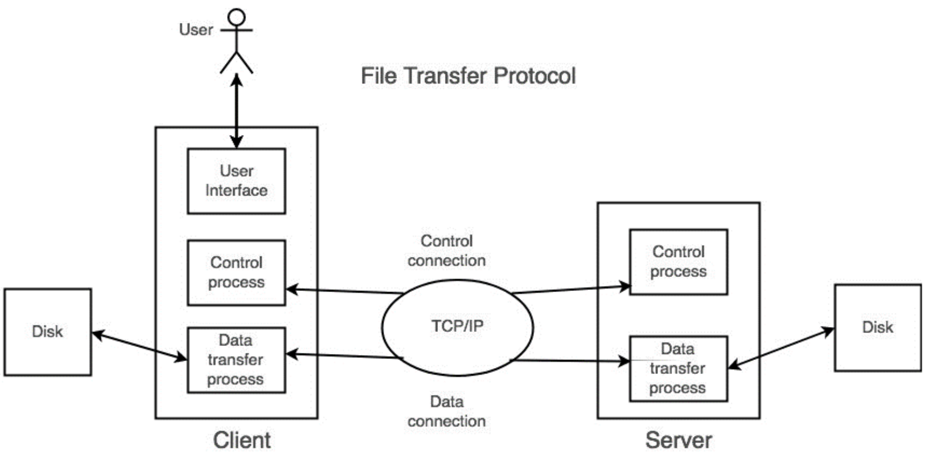

Figure 1 File Transfer Protocol

What is control connection?

FTP uses a control link to send control information such as user identity, passwords, commands to modify the remote directory, commands to retrieve and store files, and so on. On port number 21, the control link is created.

What is data connection?

FTP allows use of a data link to transfer the entire request. On port number 20, a data link is created.

Since FTP uses a different control link, the control information is sent out-of-band. Some protocols use the same TCP link to transmit their request and response header lines, as well as the data. They are said to give their control details in-band as a result of this. HTTP and SMTP are two examples of this.

FTP Session

When a client and server have an FTP session, the client establishes a control TCP connection with the server. Over this, the client sends control detail. When the server receives this information, it establishes a data link with the client. Over a single data link, only one file can be sent. The control link, on the other hand, is kept active during the user session. HTTP is stateless, which means it does not need to keep track of any user state. However, FTP must keep track of the user's status during the session.

Data Structures

FTP supports three different kinds of data structures:

File Structure – There is no internal structure of file-structure, and the file is treated as a single series of data bytes.

Record Structure – A file with a record structure is made up of consecutive entries.

Page Structure – A file with a page structure is made up of separate indexed files.

FTP Commands

Some FTP commands are as follows:

USER: The user identifier is sent to the server with this instruction.

PASS: The user password is sent to the server with this instruction.

CWD: This command helps the user to store or retrieve files in a separate directory or dataset without changing his username or accounting records.

RMD: The directory listed in the path-name is deleted as a directory with this instruction.

MKD: This command creates a directory in the directory defined by the pathname.

PWD: The name of the current working directory is returned in the response from this order.

RETR: This command instructs the remote host to establish a data link and transfer the requested file over it.

STOR: This order triggers a file to be saved in the remote host's current directory.

LIST: Sends a document for a list of all the files in the directory to be shown.

ABOR: This order instructs the server to terminate the previous FTP service command and all data transfers associated with it.

QUIT: This order terminates a USER, and the server closes the control connection if no file transfer is in progress.

FTP Responses – Some of the FTP responses are as follows:

200 Command okay.

530 Not logged in.

331 User name okay, need a password.

225 Data connection open; no transfer in progress.

221 Service closing control connection.

551 Requested action aborted: page type unknown.

502 Command not implemented.

503 Bad sequence of commands.

504 Command not implemented for that parameter.

Anonymous FTP

On certain pages with publicly accessible files, anonymous FTP is allowed. These files can be accessed without the use of a username or password. Instead, by contrast, the username is anonymous and the password is visitor. User access is severely restricted here. The user can be able to copy files but not move through folders, for example.

12.3 E-mail

Electronic mail (e-mail) is one of the most used Internet services. This service enables an Internet user to send a formatted message (mail) to another Internet user located anywhere in the world. Messages in the mail contain not only text, but also photographs, audio, and video files. The person who sends mail is known as the sender, and the person who collects mail is known as the receiver. It's similar to the postal mail service.

Components of E-Mail System:

The following are the main components of an email system: User Agent (UA), Message Transfer Agent (MTA), Mail Box, and Spool file. This are clarified in the following paragraphs:

User Agent(UA): The UA is usually a service that sends and receives mail. It is also known as a mail reader. It supports a wide range of commands for sending, receiving, and replying to messages, as well as manipulating mailboxes.

Message Transfer Agent(MTA):

MTA is in charge of transferring mail from one device to another. A device must have both a client MTA and a system MTA in order to send email. If the receivers are wired to the same machine, it sends mail to their mailboxes. If the destination mailbox is on another node, it sends mail to a peer MTA. Simple Mail Transfer Protocol is used to deliver messages from one MTA to another as seen in Figure 2.

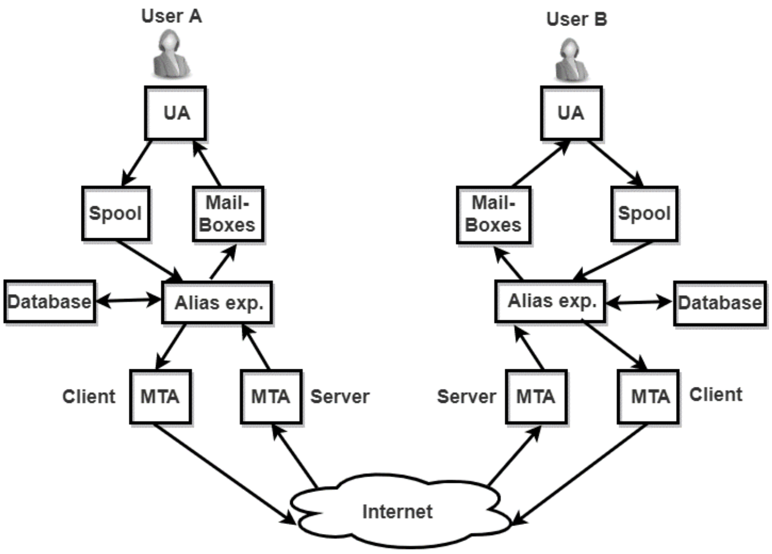

Figure 2 Message Transfer Agent

Mailbox

It's a local hard drive disc that collects e-mails. This file contains delivered emails. The user can read it or delete it depending on his or her needs. Each customer must have a mailbox in order to use the e-mail system. Just the mailbox user has access to the mailbox.

Spool File

This file holds all of the emails that need to be sent. SMTP is used by the user agent to append outgoing mails to this register. For distribution, MTA removes pending mail from the spool register. In e-mail, a single name, known as an alias, may be used to describe several e-mail addresses.

Whenever a user has to send a post, the machine checks the name of the receiver against the alias database. If a mailing list is present for a given alias, separate messages must be prepared and handed to MTA, one for each entry in the list. If no mailing list exists for the given alias, the name becomes the identifying address, and a single letter is sent to the mail transfer individual.

Services provided by E-mail system:

Composition:

The term "composition" refers to the method of creating messages and responses. Every kind of text editor may be used to compose.

Transfer

The sending of mail from the sender to the receiver is referred to as a transfer.

Reporting

The term "reporting" applies to the announcement of postal delivery. It allows users to see if their mail has been delivered, lost, or rejected.

Displaying

It applies to current mail in a format that the user can comprehend.

Disposition

This move is concerned with the recipient's behaviour after receiving mail, such as saving it, deleting it before reading it, or deleting it after reading it.

12.4 POP and IMAP3

Both POP3 (Post Office Protocol 3) and IMAP (Internet Message Access Protocol) are MAA (Message Accessing Agent) protocols that are used to retrieve messages from the mail server and deliver them to the recipient's device. Spam and virus filters are aware of all of these protocols. POP3 is more rigid and complex than IMAP.

Difference Between POP3 and IMAP: Table 4 Difference between POP3 and IMAP.

Table 4 Difference between POP3 and IMAP

Post Office Protocol	Internet message access protocol (IMAP)
POP is a straightforward protocol that allows you to download messages from your Inbox to your machine.	IMAP is a more sophisticated protocol that helps you to see all of the folders on the mail server.
On port 110, the POP server listens, and on port 995, the POP with SSL safe (POP3DS) server listens.	On port 143, the IMAP server listens, and on port 993, the IMAP with SSL secure(IMAPDS) server listens.
POP3 allows you to view your email from just one computer at a time.	Messages are accessible from a variety of platforms.
It is necessary to download the mail on the local machine in order to read it.	Before downloading the message, you can read it in half.
The recipient cannot organise his or her emails in the mail server's mailbox.	The emails can be organised directly on the mail server by the recipient.
On the mail server, the user cannot make, erase, or rename emails.	On the mail server, the user will build, erase, and rename emails.
Before uploading mail to the local system, the user cannot browse the content.	Until uploading, a user can scan the content of an email for a certain string.

There are two modes available: delete and keep. After retrieval, the mail is removed from the mailbox in delete mode. The mail is kept in the mail box after retrieval in keep mode.	Multiple backup backups of the letter are stored on the mail system, so that even if a local server's message is lost, the mail will still be recovered.
Local email tools can be used to make changes to the mail.	Online interface or email programme changes are synchronised with the server.
Many of the messages are downloaded at the same time.	Prior to uploading, the message header can be accessed.

12.5 Domain Name System(DNS)

DNS is a service that converts a host's name to an IP address. The Domain Name System (DNS) is a hierarchical network that is applied as a hierarchy of name servers. It's an application layer protocol that allows clients and servers to send and receive messages.

Requirement

A host is known by its IP address, but people have a hard time recalling numbers, and IP addresses are not static, so a mapping is needed to convert a domain name to an IP address. As a result, DNS is used to transform a website's domain name to a numerical IP address.

Domain

There are many types of DOMAIN:

1. ***Generic Domain***: All of these are common domains:.com (commercial),.edu (educational),.mil (military),.org (non-profit organisation), and.net (similar to commercial).
2. ***Country domain*** .in (india) .us .uk
3. ***Inverse domain*** if we want to know what is the domain name of the website. Ip to domain name mapping. So, DNS can provide both the mapping for example to find the ip addresses of geeksforgeeks.org then we have to type nslookup www.geeksforgeeks.org.

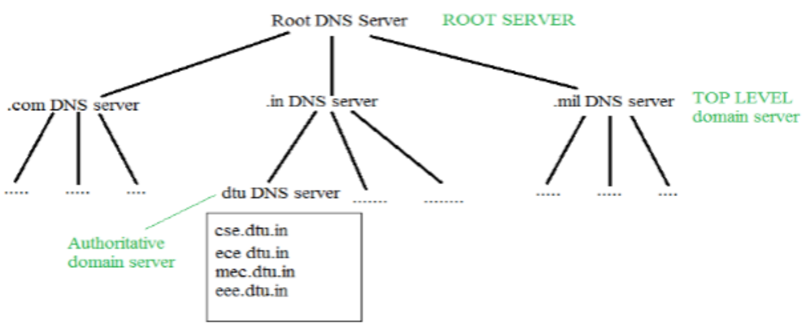

Figure 3 Organization of Domain

It is very difficult to determine the IP address associated with a website and there are millions of them. With all of those websites, we should be able to produce the IP address almost instantly; there should be no significant delay. Database organisation is critical.

DNS Record

What is the validity of a domain name and an IP address? What time is it to live? as well as other details pertaining to the domain name These documents are organised in a tree-like format.

Namespace

A list of titles that can be either flat or hierarchical. A naming scheme retains a list of name-to-value bindings – given a name, a resolution function returns the value that corresponds to it.

Name Server

It's a resolution process that's been put into action. DNS (Domain Name System) is an Internet name service. A zone is an administrative entity, and a domain is a subtree.

Name to Address Resolution

The host requests that the domain name be resolved by the DNS name server. The name server then returns to the host the IP address that corresponds to that domain name, allowing the host to bind to that IP address in the future.

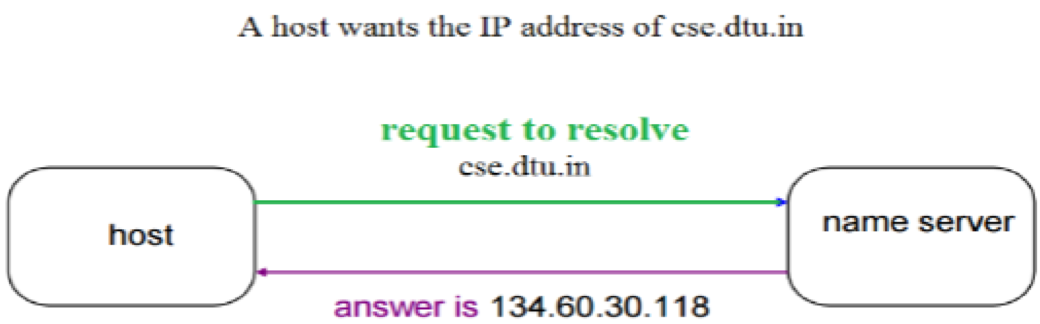

Figure 4 Name Address Resolution

Hierarchy of Name Servers

Root name servers: Name servers that are unable to address the name touch it. If the name mapping is unknown, it contacts the authoritative name server. The mapping is then obtained, and the IP address is returned to the host.

Top level server: It's in charge of com, org, and edu, as well as all top-level country domains like uk, fr, ca, and in. They have information on authoritative domain servers and know the names and IP addresses of each second-level domain's authoritative name server.

Authoritative name servers: This is the DNS registry for the enterprise, and it provides authoritative hostname to IP routing for the organization's servers. It can be kept up to date by a company or a service provider. To get to cse.dtu.in, we must first query the root DNS server, which will then direct us to the top-level domain server, and finally to the authoritative domain name server, which contains the IP address. As a result, the associative ip address will be returned by the authoritative domain server.

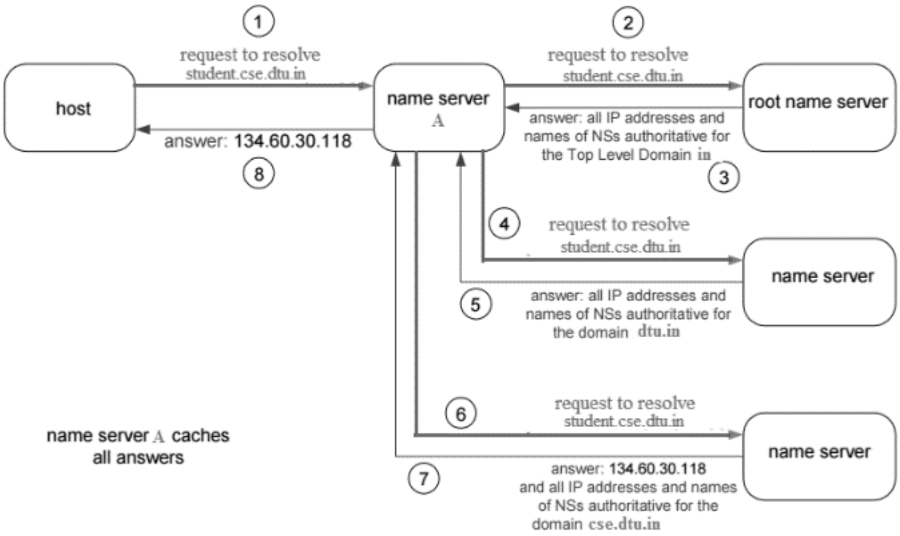

Figure 5 Domain name server

The client computer sends a request to the local name server, which, if root cannot locate the address in its database, sends a request to the root name server, which then routes the request to an intermediate or authoritative name server. Any hostName to IP address mappings can also be found on the root name server. The definitive name server is still known to the intermediate name server. Finally, the IP address is returned to the local name server, which then forwards it to the host.

12.6 World Wide Web (WWW)

The World Wide Web, also known as the web, is abbreviated as WWW. In 1989, CERN (European Library for Nuclear Research) launched the World Wide Web.

History

It's a project started in 1989 by Timothy Berner's Lee to help CERN researchers collaborate more efficiently. The World Wide Web Consortium (W3C) is a non-profit organisation dedicated to furthering web growth. Tim Berners-Lee, dubbed the "Father of the Internet," is in charge of this organisation.

System Architecture

From the user's perspective, the internet is a large, global network of documentation or web sites. Each page can provide links to other sites on the internet. The pages can be retrieved and accessed using a variety of browsers, including Internet Explorer, Netscape Navigator, Google Chrome, and others. The browser retrieves the requested page, interprets the text and formatting commands on it, and shows the page on the screen, correctly formatted.

The fundamental model of how the internet operates is shown in the Figure 6 below. On the client computer, the browser is viewing a web address. When a user clicks on a line of text that links to a page on the abd.com site, the browser follows the hyperlink by requesting the page from the abd.com server.

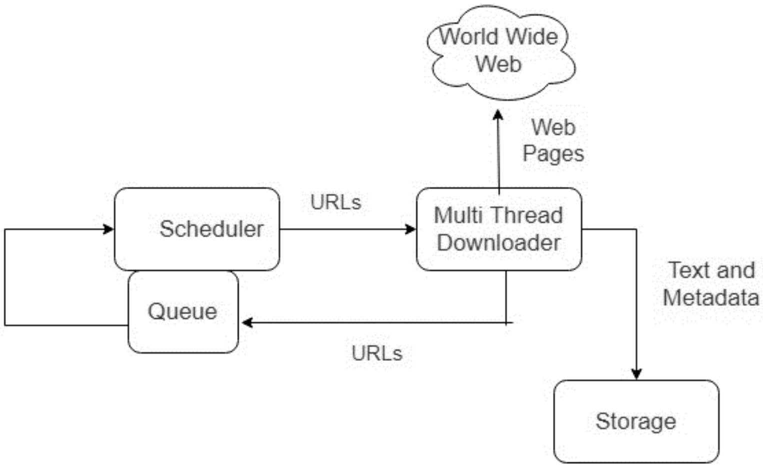

Figure 6 Basic model of web

The browser is now viewing a web page from the client computer. When a user clicks on a line of text that links to an abd.com website, the vbrowser supports the hyperlink by submitting a request to the abd.com server for the page.

Working of WWW

Web servers, Hypertext Markup Language (HTML), and Hypertext Transfer Protocol (HTTP) are among the technologies that make up the World Wide Web (HTTP).

To view webpages, you'll need a Web server. Web browsers are applications that use the Internet to view text, documents, images, animation, and video.

Web browsers provide a software interface for accessing hyperlinked resources on the World Wide Web. Initially, Web browsers were mainly used for browsing the Internet, but they have since been more widely used. Web browsers can be used for a variety of functions, including searching, mailing, and uploading files, among others. Internet Explorer, Opera Mini, and Google chrome are some of the most popular browsers.

Features of WWW

- HyperText Information System
- Cross-Platform
- Distributed
- Open Standards and Open Source
- Uses Web Browsers to provide a single interface for many services
- Dynamic, Interactive and Evolving.
- "Web 2.0"

Components of Web

The site is made up of three parts:

Uniform Resource Locator (URL): serves as a web-based resource management system.

Hypertext Transfer Protocol (HTTP): specifies communication of browser and server.

Hyper Text Markup Language (HTML): defines structure, organization and content of webpage.

12.7 Hypertext transfer protocol

The HyperText Transfer Protocol (HTTP) stands for HyperText Transfer Protocol. Tim Berner is the one who came up with the idea. HyperText is a particular category of text that is coded using a common coding language known as HyperText Markup Language (HTML). HTTP/2 is the most recent version of HTTP, released in May 2015. HyperText Transmission Protocol is a collection of protocols for transferring hypertext between two computers.

HTTP establishes a networking protocol between a web browser and a web server. It's a series of guidelines for moving data from one device to another. On the World Wide Web, data such as text, photographs, and other multimedia files are exchanged. When a computer user opens their tab, they are inadvertently using HTTP. It's an implementation protocol for hypermedia knowledge systems that are distributed and interactive.

How it works

To begin, if we want to access a website, we must first open a web browser and then enter the website's URL (e.g., www.facebook.com). This URL has now been forwarded to the Domain Name Server (DNS). The DNS server will then search their cache for records for this URL, and then return an IP address to the web browser that corresponds to this URL. The browser will now send requests to the actual server as seen in Figure 7.

The link will be closed until the server has sent data to the device. If we want something different from the server, we must re-establish the relationship between the client and the server.

The connection will be ended once the server has sent data to the client. If we want anything different from the server, we must re-establish the connection between the client and the server.

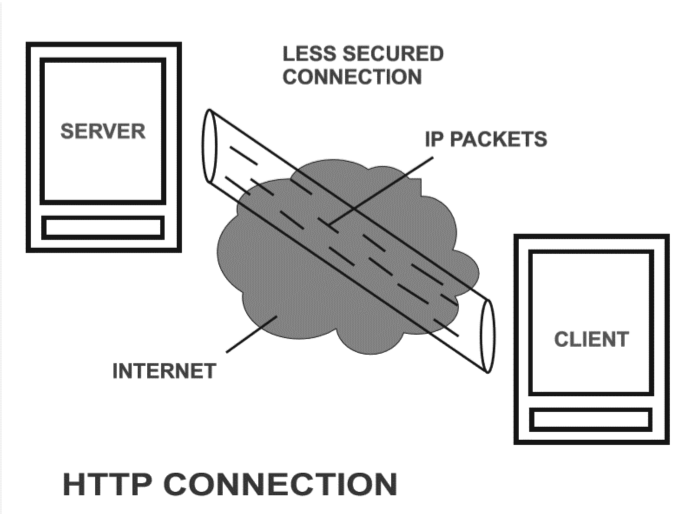

Figure 7 HTTP Connection

History

Tim Berners-Lee and his CERN team are credited with creating HTTP and related technologies.

1. HTTP version 0.9 –

This was the original version of HTTP, released in 1991.

2. HTTP version 1.0 –

RFC 1945 (Request For Comments) was included to HTTP version 1.0 in 1996.

3. HTTP version 1.1 –

RFC 2068 was introduced in HTTP version 1.1 in January 1997. In June 1999, RFC 2616 was published, which included improvements and modifications to the HTTP version 1.1 standard.

HTTP version 2.0 –

On May 14, 2015, the HTTP version 2.0 standard was published as RFC 7540.

HTTP version 3.0 –

Version 3.0 of HTTP is based on an earlier RFC draught. It has been renamed HyperText Transfer Protocol QUIC, which is a Google-developed transport layer network protocol.

Characteristics of HTTP:

- HTTP is an IP-based communication protocol for transferring data from a server to a client and vice versa:
- The server handles a client request, and the server and client are only aware of each other for the present request and response time.
- Any sort of data may be transmitted as long as the server and client are both compliant.
- Once data has been shared, the servers and clients are no longer linked.
- It's a client-server-based request-and-response protocol.
- It is a connection-less protocol because the server does not remember anything about the client when the connection is ended, and the client does not remember anything about the server.
- It's a stateless protocol since neither the client nor the server expects anything from the other, yet they can still interact.

Advantages

- Because there are fewer simultaneous connections, memory and CPU utilisation are minimal.
- Because there are fewer TCP connections, network congestion is reduced.
- Since handshaking is done at the beginning of the connection, latency is minimised because future requests do not require handshaking.
- Reports without disconnecting the connection might be the cause of the problem.
- HTTP provides request or response pipelining.

Disadvantages

- To establish communication and transport data, HTTP demands a lot of electricity.
- HTTP is less secure, because it does not employs any encryption mechanism like https utilises TLS to encrypt typical http requests and answer.
- HTTP is not suited for mobile devices and is excessively chatty.
- HTTP does not allow for true data exchange since it is insecure.
- Clients do not stop connections until they have received all of the data from the server; as a result, the server must wait for the data to be completed and is unavailable to new clients during this period.

Summary

- The Domain Name System (DNS) allows for rapid translation of IP address text from a directory of billions of addresses in a fraction of a second. Domain ideas, which employ hierarchical structures of text addresses translation, might make this possible. The servers that keep track of addresses are dispersed throughout the globe.
- To send data, HTTP employs the TCP transport service via sockets. The HTTP client establishes a TCP connection with the HTTP server by utilising sockets on port 80. The server responds to client queries with HTML pages and objects after accepting the connection from the client. HTML pages and other objects are therefore sent back and forth between the client browser and the web server.
- The World Wide Online, or Web, is an information system in which documents and other web resources are identified by Uniform Resource Locators (URLs), which may be connected together via hyperlinks, and are accessible over the Internet.

- Electronic mail is one of the most widely used network services, and it employs a user agent and a message transfer agent to transmit messages from a user's inbox to remote mailboxes. Websites have been given a new lease on life thanks to multimedia apps, which have made them more dynamic. The amalgamation of many media such as text, images, video, and sound into a single medium has made a significant contribution.
- The File Transfer Protocol (FTP) is a standard communication protocol for transferring computer files over a computer network from a server to a client. FTP is based on a client–server architecture, with the client and server having independent control and data connections.
- Electronic mail (often known as e-mail) is a technique of sending and receiving messages ("mail") between persons who use electronic equipment.

Keywords

Browser: A browser is a piece of software that your computer uses to access the Internet and view WWW content.

Domain Name System (DNS): It is responsible for defining the protocol that allows clients and servers to interact with one another. DNS allows a system to employ a resolver, which converts the host name to an IP address that the server can comprehend.

Electronic mail: It refers to the electronic form of postal mail that employs a user agent and a message transfer agent to deliver the message to the appropriate mailbox. Multimedia

HTTP: HTTP (Hypertext Transfer Protocol) is a network protocol that is used to visit any website.

World wide web (WWW): The World Wide Web (WWWW) is a system for displaying text, graphics, and audio that has been downloaded from the internet. The content and hyperlinks in a hypertext page are written in HyperText Markup Language (HTML), and the page is given an Internet address called a Uniform Resource Locator (URL) (URL).

E-mail: E-mail, in full electronic mail, messages transmitted and received by digital computers through a network.

FTP: Files are either uploaded or downloaded to the FTP server when you provide them over FTP. The files are transmitted from a personal computer to the server when you upload them. The files are transmitted from the server to your own computer when you download them.

POP: For transmitting messages from an e-mail server to an e-mail client, the post office protocol (POP) is the most widely used message request protocol on the Internet. POP3 is an e-mail protocol in which the client requests new messages from the server, and the server "pops" all new messages to the client.

IMAP: The Internet Message Access Protocol (IMAP) is an Internet standard protocol for retrieving email messages from a mail server via a TCP/IP connection by email clients. RFC 3501 is the standard that defines IMAP.

Review Questions

1. What is FTTP?
2. Write advantages and disadvantages of HTTP.
3. Write down components and features of WWW.
4. Write down most commonly used telnet commands.
5. Explain E-mail. Write down components of E-mail.
6. What is domain name system?
7. Write down hierarchy of DNS.
8. Write down difference between POP3 and IMAP.

Self-Assessment

1. The default connection type used by HTTP is _________
   a. Persistent
   b. Non-persistent
   c. Can be either persistent or non-persistent depending on connection request
   d. None of the mentioned
2. The HTTP request message is sent in _________ part of three-way handshake.
   a. First
   b. Second
   c. Third
   d. Fourth
3. The first line of HTTP request message is called _____________
   a. Request line
   b. Header line
   c. Status line
   d. Entity line
4. Dynamic web page ______
   a. is same every time whenever it displays
   b. generates on demand by a program or a request from browser
   c. both is same every time whenever it displays and generates on demand by a program or a request from browser
   d. is different always in a predefined order
5. What is a web browser?
   a. a program that can display a web page
   b. a program used to view html documents
   c. it enables user to access the resources of internet
   d. all of the mentioned
6. Expansion of FTP is __________
   a. Fine Transfer Protocol
   b. File Transfer Protocol
   c. First Transfer Protocol
   d. Fast Transfer Protocol
7. The data transfer mode of FTP, in which all the fragmenting has to be done by TCP is ________
   a. Stream mode
   b. Block mode
   c. Compressed mode
   d. Message mode
8. Which of these is not a medium for e-mail?
   a. Intranet
   b. Internet
   c. Extranet
   d. Paper
9. If the sender wants an option enabled by the receiver, it sends a ______ command.
   a) WILL
   b) DO

   c) WONT
   d) None of above

10. FTP uses the srvices of
   a) UDP
   b) IP
   c) TCP
   d) None of above
11. Which of these do not provide free E-mail?
   a. Hotmail
   b. Rediff
   c. WhatsApp
   d. Yahoo
12. Telnet protocol is used to establish a connection to __________
   a. TCP port number 21
   b. TCP port number 22
   c. TCP port number 23
   d. TCP port number 25

Answers

1.	a	2.	c	3.	a	4.	b
5.	d	6.	b	7.	a	8.	d
9.	b	10.	c	11.	c	12.	c

Further Readings

Andrew S. Tanenbaum, *Computer Networks,* Prentice Hall.

Behrouz A. Forouzan, Sophia Chung Fegan, *Data Communications and Networking,* McGraw-Hill Companies.

Burton, Bill, *Remote Access for Cisco Networks,* McGraw-Hill Osborne Media.